Jingkun HOU

AI/ML Engineer | Multimodal LLMs | Computer Vision | NLP | RAG | Pipeline Evaluation

Tel: +46 769740614 | Email: KarlJingkun@outlook.com | Github: Karl-7 | Linkedin: jingkun-hou-5250b5291 | Stockholm, Sweden

PROFILE

AI/ML Engineer specializing in **Multimodal data processing**, **LLM post-training, SFT**, and Agentic pipeline construction. **Double Master's** graduate from **KTH Royal Institute of Technology** and **EURECOM Research Institute**, with industrial experience at the **Swedish National Archives (Riksarkivet)** and **Ericsson R&D**.

EDUCATION

- **KTH Royal Institute of Technology** (Double Degree Master) — Aug. 2024 - Sep. 2026
  Master of Science: ICT Innovation - Autonomous Systems and Intelligent Robots
- **EURECOM** (EIT Double Degree Master) — Aug. 2024 - Sep. 2026
  Master of Science: Intelligence and communication system
- **University of Science and Technology Beijing**
  Bachelor of Engineering: Artificial Intelligence — Sep. 2020 - Jun. 2024
  Exchange Year - De Montfort University (2022-2023)

WORK EXPERIENCE

- **Swedish National Archives (Riksarkivet)** — Feb. 2026 - Sep. 2026
  Research Assistant - AI Lab
  Architecting and fine-tuning **multimodal VLM/LLM**s for **large-scale document analysis** and understanding. Developing and evaluating **MLLM/VLM**-based OCR pipelines with both structure-wise modification and post-training.
- **ERICSSON** — Jun. 2025 - Sep. 2025
  AI Intern in R&D
  Investigated **Reinforcement Learning (RL)** for runtime verification in **network throughput optimization**. Developed a DQN-based solution with ns-3 (network simulator) to optimize data offloading for vehicles under dynamic conditions.
- **Beijing UPhoton Technology Development**
  Deep Learning Engineer R&D — Oct. 2023 - Nov. 2023
  Responsible for comparing **object detection** networks based on PyTorch, data labeling and preprocessing.

KEY PROJECTS & PUBLICATIONS

- ***Evaluating Large Vision Language Models for Optical Character Recognition of Historical Newspapers***
  Master Thesis Project — Feb. 2026 - Sep. 2026
  Investigated VLM-based OCR for historical newspapers. Extended and **fine-tuned Multimodal VLM** with **s**tructural and spatial information retrieval, and evaluated the resulting **multimodal pipeline** against state-of-the-art OCR systems.
- ***Building AI-Powered Datasets: Mastering Multitask Learning for Next-Level Computer Vision***
  Semester Project with PhD Team — Nov. 2026 - Jan. 2026
  Investigated, implemented, and evaluated state-of-the-art **Super Resolution** and **Image Caption** algorithms for data generation of next level JPEG AI with open source datasets.
- ***Research and Application of Target Segmentation and Tracking for Material Microscopic Image Sequences***
  Bachelor Thesis Project — Dec. 2023 - Jul. 2024
  Responsible for developing **Deep Learning (DL)** networks for **Video Object Segmentation (VOS)**, Developing end-to-end pipeline for sequence image data for 3D-reconstruction. Project recognized as an **excellent thesis of the school**.
- ***Ensemble Simplified ResNet Models Based on Egyptian Hieroglyphs Data Set*** — Sep. 2023 - Oct. 2023
  DOI: 10.1109/EIECS59936.2023.10435471
  Responsible for **image data preprocessing** and **Deep Learning (DL) OCR model** structure modification, data collation and paper publishing. Contact Author of the manuscript:.

TECHNICAL SKILLS

AI & Computer Vision:	VLMs/LLM fine-tuning/post-training, image processing, Code Review, RAG+Vector Database, DL/RL/ML.
Programming:	Python (PyTorch), C++, C, Linux, Git ; Data processing, LangGraph, reAct.
Robotics:	Control algorithms, Embedded systems , Telecommunication and Networks, multi-agent communication.
Language:	English: Fluent, Swedish: A2, German: B1, French A2, Chinese: Native